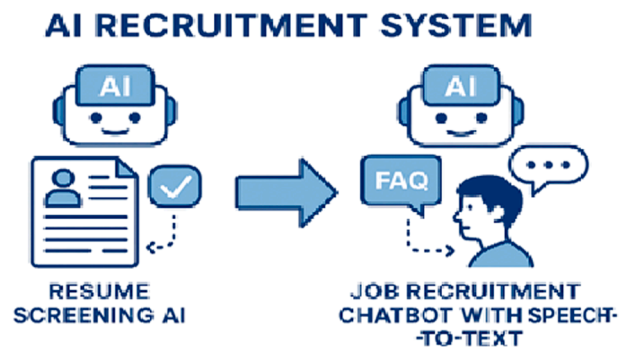


Resume data extract and job recruitment Chatbot features for AI-based resume screening & analytics

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GRAPHICAL ABSTRACT



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ABSTRACT

AI technologies are changing the field of manpower recruitment since they make it much more efficient, accurate, and scalable than conventional approaches. The project builds an AI recruitment system that is combined to have two main elements of an integrated AI recruitment system including a Resume Screening AI and a Job Recruitment Chatbot, which have the goal of improving the process in hiring as well as making the experience of the candidates better in the process.

The Resume Screening AI uses a mixed approach incorporating both classical document processing, and capabilities of advanced language models. The unstructured data of raw resumes are mined and normalized into standard forms to allow evaluation and ranking of the candidates in an organized and systematic manner according to the position's requirements. The Job Recruitment Chatbot entails a programmed chat system of interactive communication during the job

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recruitment procedure comprising the component of FAQ, conversation-based direction, and voice-to-text dynamic to make the system more accessible to a diverse group of users.

- **Document Processing Pipeline:** Parsed all-format resumes by the means of PyPDF2 and python-docx libraries and programmed the data in a structured manner, via Google Gemini 1.5 Flash API and engineered special prompts to validate the set of JSON-Schema.
- **Intelligent Screening System:** Created automated candidate screening based on (large language model) inference process to compare resume text with job requirements, producing relevance scores and classified evaluations.
- **Interactive Chatbot Development:** Developed natural language processing AI interface with chat capabilities and with speech-to-text and FAQ automation that could be used to answer candidate questions and optimize the recruitment process.

Methods

Such an integrated system proves to have immense capability in minimizing manual work load and accuracy of a decision and the general efficacy in the recruitment process of an organization at different scales.

Specifications table

Subject area	Computer Science
More specific subject area	Data Science
Name of your method	Fine-Tune Whisper-Large, TF-IDF and BERT for Question-and-Answer Validator
Name and reference of original method	Fine-Tune Whisper-Large, Fine-Tune BERT
Resource availability	https://www.openslr.org/12

Background

The global digital transformation has seen artificial intelligence (AI) transform many industries like recruitment and analytics. Nevertheless, several issues remain in the way of divine process in the resume screening and recruitment [1].

Moreover, Manual resume Data Extraction is still a time problematic, error prone task. To fill position during the recruitment process there is a big volume of resumes to be filtered, identified and analyzed to candidates’ qualification, skills and experience. This typically delays and is inaccurate, so the ones that are critical information that identify the best candidates may go missing [2].

In addition, interaction modes in job recruitment chatbots are severely limited and hinder inclusivity and user experience. Existing chatbots mostly enable text-based interaction, and candidates who are physically impaired, or have limited language proficiency, may be disadvantaged. Adding a speech to text (STT) function can greatly boost accessibility and comfort which can make a much wider field of candidates using it without a hitch [3].

Accuracy of STT technology poses another challenge. Transcription errors during interaction with a chatbot occur when incoming messages have variations in accents, dialects, speech speeds and background noise, reducing reliability of the chatbot. Consider that it’s particularly important that these aren’t inaccurate in recruitment cases where the precise answers of candidates are essential so they can judge fairly and effectively [4,5].

Addressing these challenges would enable the efficiency, accessibility, and accuracy of AI based recruitment solutions, making the way organizations screen candidates and make hires [6].

Method details

System architecture overview

The AI-powered hiring system consists of four interconnected modules: (1) Resume Extraction (Gemini 1.5 Flash) [7], (2) Speech-to-Text (Whisper-Large) [8–10], (3) Q&A Validation (TF-IDF and BERT) [11,12], and (4) Chat Interface (RASA) [13–16]. Each module functions independently while supporting the overall process.

Resume data extraction module

• Text Extraction Process

The resume pipeline starts with ingesting resumes (PDF, DOC, DOCX) and extracting raw text using OCR and parsing libraries,

handling both structured and unstructured formats [17].

• Gemini 1.5 Flash Integration

After text extraction, a standardized prompt is generated for the Gemini 1.5 Flash model to extract key resume details (e.g., personal info, education, experience) in JSON format for further processing [7].

• Data Validation and Error Handling

JSON files are parsed and validated for format and completeness. Errors triggers retry routines or flag resumes for human review, while valid data is stored in the system database with appropriate metadata [17–19].

Speech-to-Text processing module

• Dataset Preparation and Audio Input Processing

Whisper-Large was fine-tuned on LibriSpeech train-clean-100 and dev-clean sets, split 90:10 for ~10,000 training and 1000 test samples. Audio was resampled to 16 kHz, with average durations of 12.9 s (train) and 6.6 s (test). Data was imported using `parse_librispeech()`, ensuring consistency and temporal variation for effective training [21–23].

• Whisper-Large Model Architecture and Configuration

The system uses OpenAI's Whisper-Large model with 24 encoder and decoder layers, fine-tuned for speech-to-text. It includes `WhisperFeatureExtractor`, `WhisperTokenizer`, and `WhisperProcessor` for processing speech and text in a sequence-to-sequence pipeline [8,20] (Fig. 1).

• Training Configuration and Fine-tuning Process

Fine-tuning used `Seq2SeqTrainingArguments` with batch size 16, learning rate 1e-5, fp16 precision, gradient checkpointing, and 500-step warmup. Training ran for 1000 steps with WER-based evaluation every 500 steps. Padding was handled by a custom collator to preserve attention masks. Losses showed stable training without overfitting [20,24] (Fig. 2, Table 1).

Question-Answer validation system

• Dataset Development and Characteristics

A custom 10,000-sample dataset was created for Q&A validation, covering 10 interview questions with ~1000 diverse responses each. Each entry includes a question, answer, and binary validity label, with a balanced split of 5000 valid and 5000 invalid samples.

• Data Preprocessing and Preparation

Data preprocessing included cleaning, binary label normalization, and stratified 80:20 train-test splitting. TF-IDF used [SEP]-joined Q&A pairs, while BERT used `bert-base-uncased` tokenizer with 128-token truncation, padding, and attention masking.

• Dual-Model Implementation and Training

TF-IDF Baseline Model: The baseline used TF-IDF with unigrams/bigrams, stopword removal, and min document frequency of 2. A logistic regression classifier with balanced class weights (liblinear solver) was trained for interpretable comparison [25,27,28].

BERT Fine-Tuning Architecture: The BERT model used `bert-base-uncased` with fp16 mixed precision, batch size 8, and a 2e-5 learning rate. Training was memory-efficient and evaluated per epoch, with the best model selected based on F1-score [11,26,29].

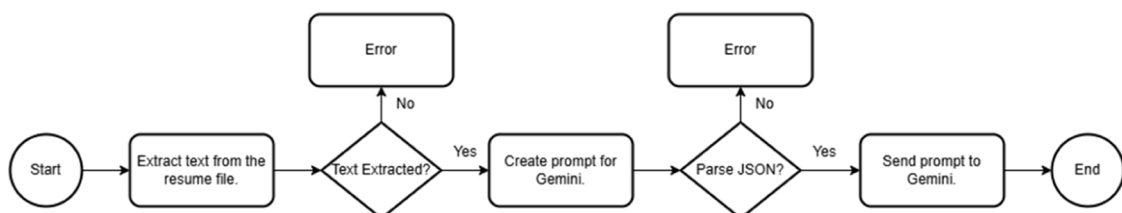


Fig. 1. Flowchart of Gemini 1.5 flash model.

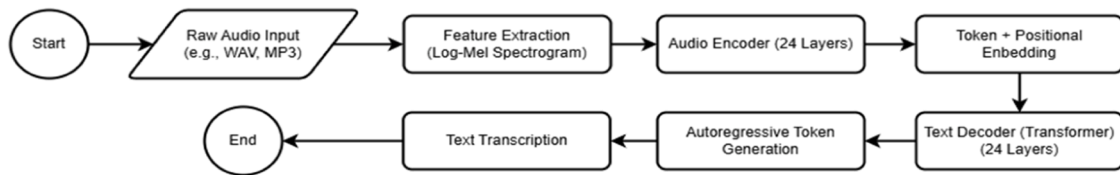


Fig. 2. Flowchart fine-tune whisper-large model.

Table 1

Training and validation loss for each step.

Step	Training Loss	Validation Loss
500	0.008600	0.112518
1000	0.006400	0.111875

• Training Results and Performance Monitoring

BERT fine-tuning showed steady improvement over 4000 steps, with training loss dropping from 0.1635 to 0.0031 and evaluation loss from 0.0973 to 0.0443. The final model was selected based on optimal F1-score, indicating effective learning without overfitting (Fig. 3).

RASA conversational interface

• Multi-Modal Input Processing

The RASA chatbot supports both text and speech inputs. Speech is transcribed by Whisper before being processed by the NLU pipeline [14].

• Natural Language Understanding (NLU)

The NLU module uses trained models to classify intents and extract entities from recruitment-specific conversations, identifying queries on jobs, interviews, and personal details [30,31].

• Dialogue Management

RASA Core manages dialogue flow and context, following predefined paths while adapting to unexpected inputs and follow-up questions [32].

• Backend Integration and Data Retrieval

The chatbot connects to the main database to access candidate and job data, enabling real-time, consistent, and dynamic responses across modules [30].

• Response Generation and Validation

Before responding, the system validates answers using trained models. Valid responses are delivered via the chatbot, while invalid ones prompt clarifications or escalate to human recruiters (Table 2).

System integration and workflow

The four modules work together: resume extraction populates the database, speech-to-text enables voice input, validation ensures

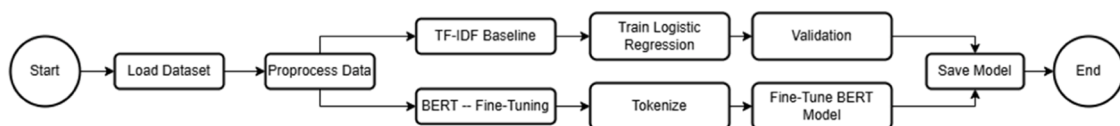


Fig. 3. Flowchart of TF-IDF and BERT for Q&A validator.

Table 2
Training Loss and Eval loss for fine-tuning BERT model.

Step	Training Loss	Eval Loss
1000	0.1635	0.0973
2000	0.0372	0.0752
3000	0.0168	0.0551
4000	0.0031	0.0443

response quality, and the RASA chatbot handles interaction. Error handling at each stage ensures system reliability and smooth user experience (Fig. 4).

Method validation

Speech recognition model performance

Speech recognition performance was evaluated using three Whisper variants on the LibriSpeech/dev-clean dataset, focusing on Word Error Rate (WER) and average transcription time to assess accuracy and efficiency.

• **Model Comparison Results**

The fine-tuned Whisper-Large model achieved the best WER at 6.03 %, outperforming Whisper-Large-v2 (6.48 %) and Whisper-Small (7.02 %). This 0.45 % improvement highlights the effectiveness of domain-specific fine-tuning.

• **Computational Efficiency Analysis**

Transcription time analysis showed a trade-off between speed and model size. Whisper-Small was fastest (0.41s/sample), while fine-tuned Whisper-Large was slower (1.61s/sample). Whisper-Large-v2 balanced both (1.43s/sample), highlighting the need to weigh accuracy against processing speed for real-time use (Fig. 5).

Question-answer validation system performance

To validate the accuracy of automated Q&A system, this project implemented and compared two distinct approaches: TF-IDF with Logistic Regression and Fine-tuned BERT, evaluated on a binary classification task (valid/invalid responses) (Fig. 6).

• **Classification Performance Metrics**

Both models performed strongly across all metrics. TF-IDF + Logistic Regression achieved 95.35 % accuracy, while fine-tuned BERT reached 99.15 %. Matching precision, recall, and F1-scores indicate balanced, unbiased classification performance.

• **Confusion Matrix Analysis**

Confusion matrix analysis confirmed both models' reliability. TF-IDF achieved strong results with 944 invalid and 963 valid correct classifications, while BERT outperformed with only 8 false positives and 9 false negatives, indicating near-perfect discrimination on 2000 samples (Fig. 7).

• **Model Selection Rationale**

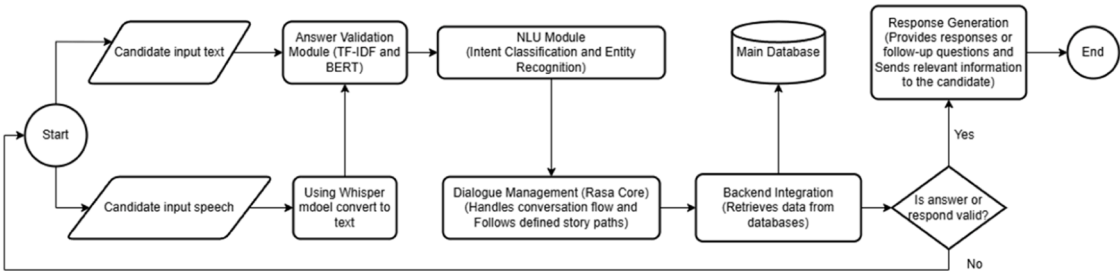


Fig. 4. Flowchart of RASA Chatbot.

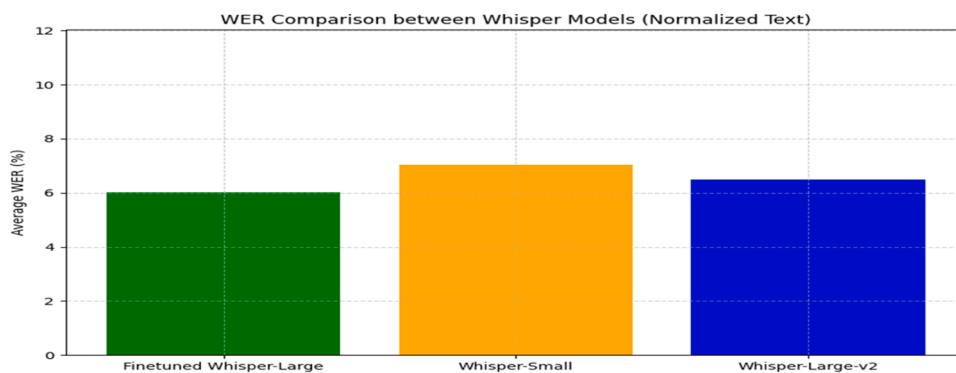


Fig. 5. Word error rate comparison between whisper model bar chart.

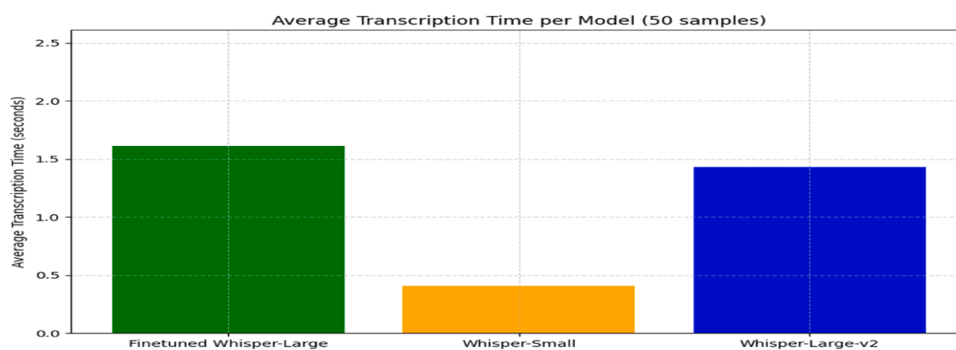


Fig. 6. Average transcription time per model bar chart.

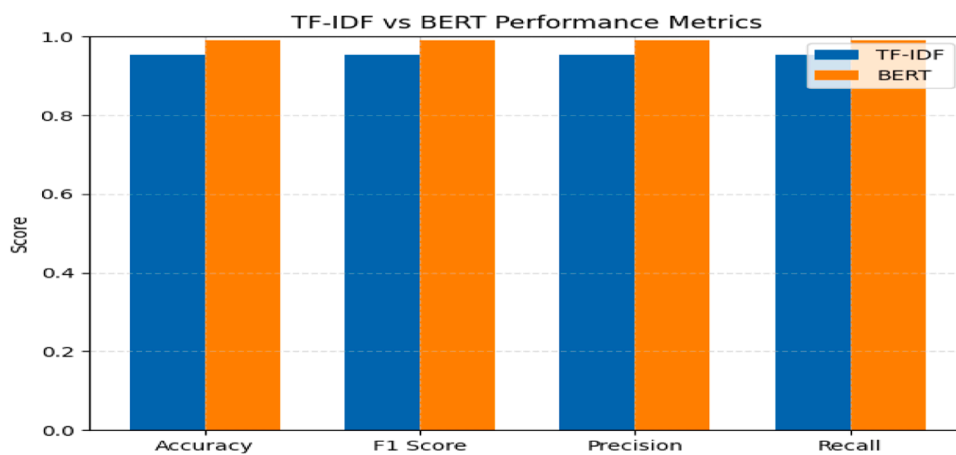


Fig. 7. TF-IDF vs BERT performance metrics bar chart.

Between the two models, the fine-tuned BERT achieved higher accuracy (99.15 % vs. 95.35 %) and more balanced error distribution, making it the preferred option for deployment. Its 3.8 % accuracy gain justifies the added computational cost through improved reliability in validation tasks (Fig. 8).

Validation framework reliability

The hybrid model registered improved performance with 93.97 % accuracy in speech recognition and 99.15 % accuracy in Q&A validation. These results validate the efficiency and operational applicability of the introduced methodology in computerized interview analysis under accuracy constraints within computationally affordable capabilities (Tables 3 and Table 4).

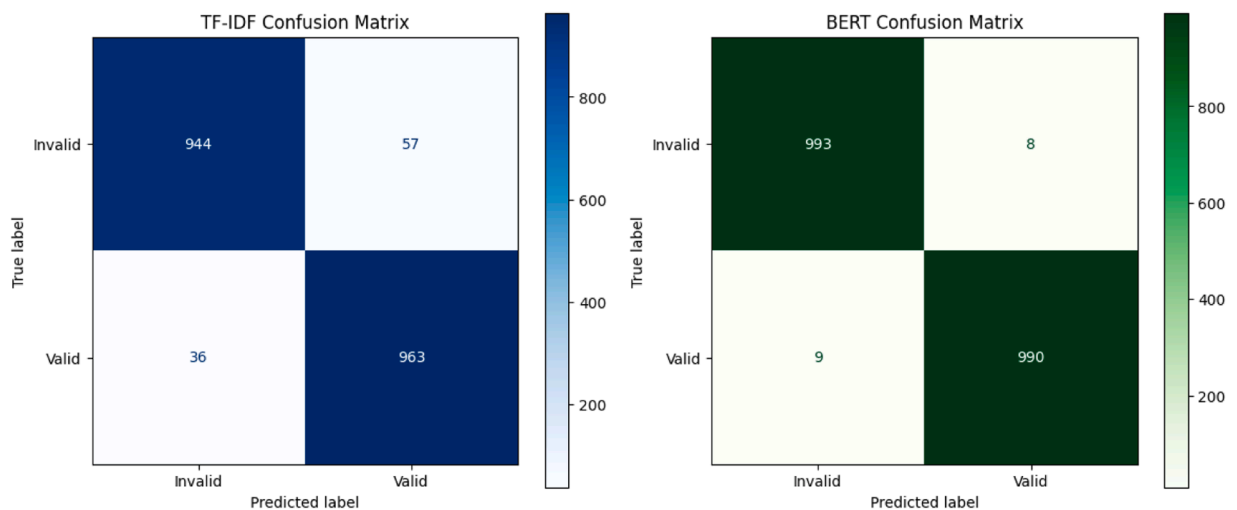


Fig. 8. Confusion matrix for TF-IDF and BERT model heatmap.

Limitations

There were system development bottlenecks. Whisper-Large training was a computationally demanding exercise, something even laptops with mediocre capability could not handle—leading to constant crashes, long training time, and space limitation. Monitoring and optimization were needed on an ongoing basis here. Creating an evenly weighted, real-world Q&A set of 10,000 samples took time, had to be populated manually to be linguistically dense and consistent. Finetuning Whisper also involved mass-scale hyperparameter tuning via trial and error to reduce Word Error Rate (WER). Integration of the custom answer validation model into Rasa architecture was also technologically demanding, particularly real-time inference synchronization with dialogue flow, latency control, and consistent validation for multi-turn conversation.

Related research article

None

For a published article

None

Ethics statements

The report does not involve human participants or data from social media platforms.

CRediT author statement

Chong Kar Weng: Research, Conceptualization, Methodology, Implementation, Writing – original draft, Writing – review & editing
 Dr. Ng Kok Why: Writing – review & editing, Supervision
 Fu Yong Hong: Writing – review & editing

Supplementary material and/or additional information [OPTIONAL]

None

Table 3

Summary table for whisper-large model result.

Model	Dataset	Word Error Rate (WER)	Avg Transcription Time
Fine-Tune Whisper-Large	LibriSpeech/dev-clean	6.03 %	1.61 s
Whisper-Large-v2	LibriSpeech/dev-clean	6.48 %	0.41 s
Whisper-Small	LibriSpeech/dev-clean	7.02 %	1.43 s

Table 4
Summary table for Q&A validator model.

Metric	TF-IDF + Logistic Regression	Fine-Tuned BERT
Accuracy	0.9535	0.9915
F1 Score	0.9535	0.9915
Precision	0.9537	0.9915
Recall	0.9535	0.9915

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I already have put in the link of one of the dataset.

References

- [1] Gan, C., Zhang, Q., & Mori, T. (2024, January 16). Application of LLM Agents in Recruitment: A novel Framework For Resume Screening. arXiv.org. <https://arxiv.org/abs/2401.08315>.
- [2] M. Blessing, AI-Powered Resume Screening: Benefits and Challenges, ResearchGate, 2025, February 5. https://www.researchgate.net/publication/388688179_AI-Powered_Resume_Screening_Benefits_and_Challenges.
- [3] A. Hirai, A. Kovalyova, Speech-to-text applications' Accuracy in English Language learners' Speech Transcription, 2024. <https://scholarspace.manoa.hawaii.edu/items/693fc7a9-b98c-4191-a1e3-b5e5b7e884d8>.
- [4] Hasan Gyulyustan, H. Hristov, S. Stavrev, Svetoslav Enkov, Measuring and analysis of speech-to-text accuracy of some automatic speech recognition services in dynamic environment conditions, AIP Conf. Proc. (2024), <https://doi.org/10.1063/5.0196448>.
- [5] K. Kuhn, V. Kersken, B. Reuter, N. Egger, G. Zimmermann, Measuring the accuracy of automatic speech recognition solutions, ACM Trans. Access. Comput. 16 (4) (2023) 1–23, <https://doi.org/10.1145/3636513>.
- [6] Li, S., Li, K., & Lu, H. (2023, July 13). National Origin Discrimination in Deep-Learning-Powered Automated Resume Screening. arXiv.org. <https://arxiv.org/abs/2307.08624>.
- [7] R. Zimbres, Multimodality With Gemini-1.5-Flash: Technical Details and Use Cases, Medium, 2024, November 23. <https://medium.com/google-cloud/multimodality-with-gemini-1-5-flash-technical-details-and-use-cases-84e8440625b6>.
- [8] Radford, A., Kim, J.W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022, December 6). Robust Speech Recognition Via Large-Scale Weak Supervision. arXiv.org. <https://arxiv.org/abs/2212.04356>.
- [9] R. Surya, A. Amrullah, Analysis of whisper automatic speech recognition performance on low resource language, Pilar Nusa Mandiri/Pilar Nusa Mandiri 20 (1) (2024) 1–8, <https://doi.org/10.33480/pilar.v20i1.4633>.
- [10] Macháček, D., Dabre, R., & Bojar, O. (2023, July 27). Turning Whisper into Real-Time Transcription System. arXiv.org. <https://arxiv.org/abs/2307.14743>.
- [11] Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2018, October 11). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv.org. <https://arxiv.org/abs/1810.04805>.
- [12] Techniques to Improve Q&A Accuracy with Transformer-based models on Large Complex Documents. (n.d.). Ar5iv. <https://ar5iv.labs.arxiv.org/html/2009.12695>.
- [13] T. Smutek, E. Golec, P. Rymarczyk, M. Jarmul, A. Hernas, Enhancing conversational ai with the Rasa framework: intent understanding and NLU pipeline optimization, J. Mod. Sci. 57 (3) (2024) 531–548, <https://doi.org/10.13166/jms/191223>.
- [14] Rafla, A., & Kennington, C. (2019, July 11). Incrementalizing RASA's Open-Source Natural Language Understanding Pipeline. arXiv.org. <https://arxiv.org/abs/1907.05403>.
- [15] Rasa Research: team, topics, & academic collaborations, Rasa. <https://rasa.com/research/>, 2022, June 3.
- [16] K.N. Lam, N.N. Le, J. Kalita, Building a Chatbot on a Closed Domain Using RASA, 118, 2020, pp. 144–148, <https://doi.org/10.1145/3443279.3443308>.
- [17] Helli, S.S., Tanberk, S., & Cavsak, S.N. (2023, June 23). Resume Information Extraction Via Post-OCR Text Processing. arXiv.org. <https://arxiv.org/abs/2306.13775>.
- [18] Cheng, Z., Zhang, P., Li, C., Liang, Q., Xu, Y., Li, P., Pu, S., Niu, Y., & Wu, F. (2022, July 14). TRIE++: Towards End-To-End Information Extraction from Visually Rich Documents. arXiv.org. <https://arxiv.org/abs/2207.06744>.
- [19] Dhouib, M., Beltaieb, G., & Shabou, A. (2023, April 24). DocParser: End-to-end OCR-free Information Extraction from Visually Rich Documents. arXiv.org. <https://arxiv.org/abs/2304.12484>.
- [20] openai/whisper-large · Hugging Face. <https://huggingface.co/openai/whisper-large>, 2023, September 13.
- [21] Librispeech: An ASR Corpus Based On Public Domain Audio Books. (2015, April 1). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/7178964>.
- [22] Openslr.org. (n.d.). <https://www.openslr.org/12>.
- [23] Papers with Code - LibriSpeech Dataset. (n.d.). <https://paperswithcode.com/dataset/librispeech>.
- [24] Openai. (n.d.). GitHub - openai/whisper: Robust Speech Recognition via Large-Scale Weak Supervision. GitHub. <https://github.com/openai/whisper>.
- [25] Oancea, B. (2025, February 27). Text Classification Using Machine Learning Methods. arXiv.org. <https://arxiv.org/abs/2502.19801>.
- [26] Text Classification with BERT Tutorial | SabrePC Blog. (n.d.). https://www.sabrepc.com/blog/Deep-Learning-and-AI/text-classification-with-bert?srsltid=AfmBOoqYV_N0o4rWhOV11uvRUuv_Hv2QtJvKEirhNtkK59YYh6w0mEd.
- [27] Lin, Y., Chen, S., Liu, J., & Lin, C. (2023, June 12). Linear Classifier: An Often-Forgotten Baseline For Text Classification. arXiv.org. <https://arxiv.org/abs/2306.07111>.
- [28] A. Ryblov, Text Classification: Baseline with TF-IDF and Logistic Regression, Medium, 2024, February 22. <https://medium.com/%40ryblovartem/text-classification-baseline-with-tf-idf-and-logistic-regression-2591fe162f3b>.

- [29] H. Amit, Fine-Tuning BERT For Classification: A Practical Guide, Medium, 2025, April 18. <https://medium.com/%40heyamit10/fine-tuning-bert-for-classification-a-practical-guide-b8c1c56f252c>.
- [30] R. Sinha, Intelligent Chatbot System based on Entity Extraction using RASA NLU, IJERT (2022), <https://doi.org/10.17577/IJERTV11IS020193>.
- [31] Bocklisch, T., Faulkner, J., Pawlowski, N., & Nichol, A. (2017, December 14). RASA: Open source Language Understanding and Dialogue Management. arXiv. <https://arxiv.org/abs/1712.05181>.
- [32] Unpacking the TED policy in rasa open source, Rasa. <https://rasa.com/blog/unpacking-the-ted-policy-in-rasa-open-source>, 2024, September 17.